

ZENABU SUBOI

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RESEARCH INTEREST

My primary research interest is in the area of infectious disease epidemiology. It is my desire to be engaged in innovative epidemiological researches and translate the research into evident-based public health practice for the control and prevention of diseases; and aid in planning and evaluating the effectiveness of public health programs.

EDUCATION

2024–Ongoing	PhD, Epidemiology , University of Basel, Swiss Tropical and Public Health Institute, Basel – Switzerland
Title	Uncertainty Quantification of Agent-Based Models with Application to Malaria Intervention Modelling
Supervisors	Dr. Christian Selinger, Dr. Emilie Pothin and Prof. Christian Lengeler
Description	This PhD seeks to quantify and propagate uncertainty from various parameter sources and to determine the impact of these uncertainties on simulating the effect of malaria interventions at the sub-national level
2019–2020	MSc., Mathematics , University of Stellenbosch, Stellenbosch – South Africa
Title	Calibrating Models To Data: A Comparison Of Methods
Supervisors	Dr. Marijn Hazelbag and Prof. Wim Delva
Description	We compared performance of model calibration methods in a simple SIR simulation setting, where Rejection ABC with an infinitesimally small tolerance was used as the reference.
2017–2018	MSc., Mathematical Science , University of Cape Town (African Institute for Mathematical Sciences (AIMS)), Cape Town – South Africa
Title	Comparing Post-sampling Correction Methods For Approximate Bayesian Computation (ABC)
Supervisors	Dr. Marijn Hazelbag and Prof. Wim Delva
Description	We compared performance of different post-sampling correction methods for ABC in a simple Agent-based model simulation setting, where likelihood estimation was used as the gold standard.

2012–2016	BSc., Mathematical Science (Statistics) , University for Development Studies, Tamale–Ghana
Title	Size Biased Exponential Distribution
Supervisor	Dr. Nasiru Suleman
Description	We proposed a size biased exponential distribution using weighted distributions

EXPERIENCE

Work Experience

- Oct. 2021– Oct. 2023** **Assistant Lecturer**, Academic City University College, Accra – Ghana
- Teach and design lesson plans for Mathematics, Data Analysis and Statistics courses
 - Record school grades and track progress from beginning of the semester
 - Prepare tests after each lesson as well as periodic quizzes to determine comprehension
 - Develop curriculum for courses for the semester
- Feb. 2018 – June 2018** **Tutor**, *GrandWest Kulani High School Saturday Maths Project*, Cape Town – South Africa.
- Taught High School students in Mathematics under the African Institute for Mathematical Sciences (AIMS-South Africa) Schools Enrichment Centre
- Sept. 2016 – Aug. 2017** **Teaching Assistant**, *University for Development Studies*, Tamale - Ghana.
- Handled tutorial lessons on Experimental Design in statistics
 - Assisted lecturers in conducting examinations and assessing students
 - Assisted some students with their undergraduate thesis

Field Work

- 2015–2016** **Internship**, *Seventh Day Adventist Hospital*, Obuasi - Ghana.
- Understudied the biostatistician in the records department
 - Collected and stored patients' data for future analyses
- 2013–2014** **Community Development**, *Third Trimester Field Practical Programme(TTFPP)*, UDS - Ghana.
- Stayed in rural community for three months in each year
 - Identified its pressing need and wrote a detailed report on it

Leadership Experience

- 2022** **STEMSheCan Mentor**, A STEM Leadership and Mentorship Advocacy Program

- Inspired, empowered and challenged young female students to participate in STEM
- Introduced young students to STEM and the possibilities and opportunities available to them

2019–2020

Welfare Secretary, Association of Ghanaian Students in Stellenbosch University (AGSSU), Stellenbosch - South Africa.

- Worked with other executives as a team
- Participated in Stellenbosch University's International Food Evening, in which we won

2015–2016

President, *National Union of Baptist Students (NUBS-UDS)*, Navrongo - Ghana.

- Worked with other executives as a team
- Took decisions on behalf of the group
- Represented the group at the Students' Christian Council meetings

Communication Experience

2018

Interviewer, AIMS - SA, Cape Town - South Africa.

- Interviewed Dr Justin Munyakazi (Senior Lecturer, University of the Western Cape) at the **15-years-of-Excellence** Generation Exchange interview at AIMS-SA, where portion of video was displayed at the August graduation that year.

ACADEMIC HONORS AND AWARDS

2019

Best Poster Pitch

2016

Overall Best Graduating Female in the Faculty of Mathematical Sciences

2016

Alumni award for Best Graduating Female Student

CONFERENCES ATTENDED

2019

International Conference on Infectious Disease Dynamics (EPIDEMICS⁷)

Location

North Charleston, South Carolina - USA

Description

A gathering of over 400 scientists where I presented a poster on "Modelling The Effect Of Mass Dog Vaccination Delivery Strategies On Time To Rabies Elimination In Rural Kenya"

2019

SACEMA Research Days

Location

Stellenbosch - South Africa

Description

Presented my project in a three-minute poster pitch and followed with a poster presentation

WORKSHOPS

2019

Introduction to R: Management, Exploration, and Communication of Data

Location

Stellenbosch University - South Africa

Description	We were taught how to manage, explore and communicate with data using tidyverse package in R
2019	Clinic on Meaningful Modeling of Epidemiological Data (MMED)
Location	Cape Town - South Africa
Description	The clinic focused on training junior researchers from the USA and Africa to conduct integrative research in infectious disease dynamics.

COMPUTER SKILLS

- R
- Python

REFEREES

Dr. Christian Selinger, Disease Modeler, Analytics and Intervention Modeling group, Swiss Tropical and Public Health Institute, Basel – Switzerland

Email – Christian.selinger@swisstph.ch

Prof. Nasiru Suleman, Head of Department, Statistics. C. K. Tedam University of Technology and Applied Sciences (CKTUTAS), Navrongo - Ghana

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Dr. Marijn C. Hazelbag, Data Scientist, Expert AI

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